A Classifier-Agnostic Bayes-Risk Benchmark for Weakly-Paired Multimodal Fusion, with AVID: A Severely Weakly-Aligned RGB-Infrared Drone Dataset

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Abstract

Every existing RGB-thermal fusion benchmark (LLVIP, M3FD, KAIST) is built on co-located or heavily calibrated rigs, so its evaluation protocol silently assumes what classical multi-view learning theory [2] requires: that the two modalities are exact, registered views of the same instance. Real dual-optic sensor rigs violate this, and every existing empirical fix reports the error of *one trained model* – confounded by architecture, training budget, and optimization luck, and telling you nothing about whether a *better* model could have done differently. We instead introduce a **classifier-agnostic Bayes-risk benchmark**: closed-form Bhattacharyya and Chernoff-information bounds [1, 3, 6] on the *optimal* classifier's error, computed directly from class-conditional feature distributions with no gradient descent anywhere. The Bhattacharyya bound is tightened by 6-13% via the more general optimized Chernoff bound, and cross-checked against a fully nonparametric kernel-density estimate computed directly on real patches (not only on synthetic self-test data) – both agree to within 11-16% and preserve the same fusion-dominates ordering. Applied to 35,952 real bird/drone instances from AVID (Aerial Visible-Infrared Drone dataset, a large-scale RGB-IR video dataset for counter-drone surveillance), this reveals a fusion advantage 3-4$\times$ larger than any trained model has ever reported for a task like this (Bhattacharyya bound at the corpus's modal 12-16px scale: RGB=0.148, IR=0.198, **fusion=0.046**); an asymmetric corruption ceiling (RGB's bound roughly triples under noise-type corruptions, IR's barely moves); and a new **Bayes Fusion Gain** metric that shrinks 16% (0.111$\rightarrow$0.093) as injected cross-modal misalignment reaches the real measured ceiling of this corpus's own misalignment (17px, from a real 300-pair sample where 32.3% of pairs exceed 5px against targets that are themselves only 8-16px wide) – instantiated alongside a numerical data-processing-inequality argument showing real mutual information between the fused representation and the label degrading by 9% over the same range. Beyond the benchmark itself, we release: 28,790 training / 3,250 validation RGB-IR pairs across 115 videos (after identifying and removing a systematic noise-augmented duplicate the source export had silently baked into every frame); AVID-C, a reproducible ImageNet-C-style corruption suite (7 families $\times$ 5 severities) used to drive the corruption-robustness ceiling above; and a measured split-leakage pitfall (naive frame-level splitting leaks 87.5% of "training" frames vs. 0.0% for the video-level splitting we use). We also disclose a real data-organization failure found during release preparation – the source Test split's RGB and IR files use non-overlapping numbering schemes with no manifest to resolve them – as a limitation rather than working around it silently. No detector is trained anywhere in this paper; every reported number is a property of the data distribution itself. Code (including everything that produced every number above) and the dataset (CC-BY-NC-4.0, Hugging Face) are both released alongside this paper.

1 Introduction

Counter-drone (anti-UAV) vision systems must detect, classify, and track small aerial objects that are adversarial to exactly the sensing modalities that are cheapest to deploy. RGB cameras fail in darkness, fog, and glare, and lose signal entirely once a drone is a few dozen pixels wide at range. Thermal (IR) cameras

keep working in the dark and see through some atmospheric scattering, but lack the spatial and textural detail needed to distinguish a drone from a bird, or to read its silhouette at all once its thermal signature is weak (e.g. a coasting, low-power drone against a warm background). The natural response – fusing RGB and IR – is well established for pedestrian and vehicle detection (KAIST [11], LLVIP [12], M3FD [14], RoadScene [19]), but the public data situation for *drones as the fused target* is structurally thinner than it looks at first glance.

The gap. We surveyed the two adjacent lines of public RGB-T and drone-vision datasets (Table 1, Figure 1) and found no dataset that is simultaneously (a) RGB-IR *paired*, (b) has a *drone* (not a pedestrian or vehicle) as the target class, and (c) ships video/tracking data rather than isolated still pairs. The RGB-T pair datasets closest in spirit to fusion research (LLVIP, M3FD, RoadScene, KAIST, FLIR ADAS) are all pedestrian/vehicle detection corpora with no drones at all. The datasets built specifically for anti-drone vision (Anti-UAV [13], Anti-UAV410 [10], the 3rd Anti-UAV Challenge [9], CST Anti-UAV [18]) are thermal-*only*: there is no paired RGB stream to fuse against. And the datasets that do pair a drone-mounted camera with thermal imagery (DroneRGBT [16], VTUAV [20]) or with RGB alone (VisDrone [21]) have the camera on the drone, looking down at pedestrians or generic scenes – the inverse configuration from anti-drone surveillance, where the camera is on the ground looking up or out at the drone.

A gap in how fusion is evaluated, not just in what data exists. The deeper problem is not merely that no dataset combines these three properties – it is that the datasets which come closest (LLVIP, M3FD, KAIST) are built on co-located or heavily calibrated rigs, so their evaluation protocols implicitly assume the two modalities are exact, registered views of the same instance. This is precisely the assumption classical multi-view learning theory [2] requires and real dual-optic sensor rigs violate. No existing RGB-T benchmark's evaluation protocol can even detect this violation, let alone measure how much fusion depends on it – and every existing empirical attempt to measure it reports the error of *one trained model*, confounded by architecture, training budget, and optimization luck, and silent about whether a different model would have told a different story. We introduce a tool that cannot have this confound because it never trains a model at all: a **classifier-agnostic Bayes-risk benchmark** (§6) built on closed-form Bhattacharyya and Chernoff-information bounds [1, 3, 6] on the *optimal* classifier's error, computed directly from real patches. Because the Bayes risk is a property of the data distribution alone, we can ask questions no single-model evaluation can answer cleanly: how does the fusion advantage scale with object size, corruption severity, or cross-modal misalignment, *in the limit of the best possible classifier*, not whatever one particular architecture and training budget happened to learn. This yields a new **Bayes Fusion Gain** metric and a numerical instantiation of the data-processing inequality [5] against real measured misalignment.

What we release. AVID is both a dataset in its own right and the testbed that makes this benchmark necessary rather than academic: RGB-IR paired video and frame-pair data with drones and birds as the target classes, captured under realistic ground-based surveillance conditions (hilly regions, forest cover, cloudy sky, fog/mist), with RGB and IR sensors that are only *weakly* aligned because that is what a real dual-optic gimbal actually delivers. Rather than pretend the pairs are pixel-registered (as LLVIP/M3FD/KAIST's rigs allow), we quantify the residual geometric offset per pair: on a real 300-pair sample, 32.3% of pairs exceed 5px of measured misalignment (§5) against targets that are themselves only 8-16px wide (§3.5) – large enough, relative to the target, that it is not a rounding error a fusion model can safely ignore.

Contributions.

1. **A classifier-agnostic Bayes-risk benchmark** (§6): closed-form Bhattacharyya and Chernoff-information bounds on the optimal classifier's error, a new **Bayes Fusion Gain** metric, and a numerical

data-processing-inequality instantiation against real measured misalignment – computed from 35,952 real patches with no model training anywhere, and applicable to any weakly-paired multimodal dataset, not only counter-drone RGB-T.

2. **AVID dataset**: large-scale, weakly-aligned RGB-IR drone/bird detection and tracking data (video + frame pairs), released on Hugging Face under CC-BY-NC-4.0, with video-level splits and per-pair alignment metadata, that makes the benchmark's misalignment sweep a measurement against a real distribution rather than an arbitrary synthetic range (§3).
3. **Weak-alignment quantification**: two independent per-pair alignment estimators (phase correlation on edge maps, and mutual-information registration search), revealing a bimodal real misalignment distribution (§5).
4. **AVID-C corruption benchmark**: a reproducible, parameterized (7 corruption families $\times$ 5 severities) extension of the original dataset's fixed-severity distortions, released as code so any subset can be corrupted at any severity, deterministically, and the perturbation channel driving §6's corruption-robustness ceiling (§4).
5. **A measured split-leakage pitfall** for short video-clip corpora: naive frame-level splitting on the real corpus leaks 87.5% of "held-out" frames vs. 0.0% for video-level splitting (§3.7).
6. **An extreme small-object characterization**: 67.6% of annotated boxes are 8-16px wide, motivating a finer-grained scale-stratified evaluation protocol than COCO's buckets, and the operating range the Bayes-risk scale sweep is evaluated over (§3.5).

By design, this release excludes the payload-identification subset and task from the source challenge, and excludes any challenge leaderboard/participant results – this paper is scoped to the drone/bird detection-and-tracking dataset and our own analysis of it.

2 Related Work

Table 1 summarizes the datasets closest to AVID along the three axes that matter for this release: whether RGB and IR/thermal are paired, whether a drone is the detection target (rather than the imaging platform), and whether the data includes video/tracking rather than isolated stills. Figure 1 positions them schematically on the first two axes.

RGB-T pedestrian/vehicle fusion. LLVIP [12] and KAIST [11] are the standard low-light pedestrian fusion benchmarks, both using co-located or beam-splitter rigs to achieve pixel alignment; M3FD [14] and RoadScene [19] target vehicle-scene fusion with similarly tight registration. FLIR ADAS is the one widely-used dataset in this group that is *not* pixel-aligned (different optics/FOV between the RGB and thermal cameras), which is closer to AVID's capture setup, but it contains no drones and is a still-image, not video, corpus.

Anti-drone vision. The Anti-UAV line of benchmarks [9, 10, 13, 18] is the closest prior art on the *target* axis – drones are exactly what these datasets track – but every one of them is thermal-only. There is no RGB stream to fuse against, so none of the RGB-T fusion techniques developed for LLVIP/M3FD-style data has a natural anti-drone benchmark to be evaluated on. AVID is, to our knowledge, the first publicly-documented dataset to close this specific gap.

Drone-view datasets. DroneRGBT [16], VTUAV [20], and VisDrone [21] all put the camera on the drone rather than pointing at one. They are valuable for aerial perception broadly, but structurally answer a different question (what does a drone see) from the one AVID targets (what does a ground-based, dual-sensor surveillance system see when it looks at a drone).

Table 1: RGB-IR paired and/or drone-vision datasets closest to AVID. "Aligned" reflects the alignment methodology reported by each dataset's authors, not a re-measurement by us. "Drone target" distinguishes datasets where a drone is the object being detected/tracked from "drone view" datasets where the camera is drone-mounted.

Dataset	RGB+IR paired	Drone target	Video/tracking	Alignment	Scale
LLVIP [12]	✓	–	–	pixel-aligned	16,836 pairs
M3FD [14]	✓	–	–	pixel-aligned	4,200 pairs
RoadScene [19]	✓	–	–	registered	221 pairs
KAIST [11]	✓	–	✓	pixel-aligned	∼95K pairs
FLIR ADAS [17]	✓	–	–	weakly aligned	26,442 frames
Anti-UAV [13]	– (IR only)	✓	✓	n/a	300+ sequences
Anti-UAV410 [10]	– (IR only)	✓	✓	n/a	410 sequences
CST Anti-UAV [18]	– (IR only)	✓	✓	n/a	220 sequences
DroneRGBT [16]	✓	– (drone view)	–	registered	∼3,600 pairs
VTUAV [20]	✓	– (drone view)	✓	well-aligned	500 sequences
VisDrone [21]	– (RGB only)	– (drone view)	✓	n/a	∼10K images
AVID (this work)	✓	✓	✓	**weakly aligned, quantified**	28,790/3,250 pairs[†]

[†]Train/Validation, deduplicated (see §3.4); a Test split exists but is not pair-matched in the current export (§3.4), so it is omitted from this row rather than reported as a pair count it isn't.

Corruption-robustness benchmarking. Our corruption suite follows the severity-graded corruption-family methodology introduced by ImageNet-C [8]. The closest existing robustness benchmark in the drone space, UAV-C [15], targets a different problem: it corrupts UAV-*view* tracking benchmarks (drone-mounted-camera footage), not anti-UAV RGB-T detection of a drone as the target. We are not aware of an existing corruption-robustness benchmark for the Anti-UAV-style drone-as-target setting, which is the gap AVID-C (§4) targets.

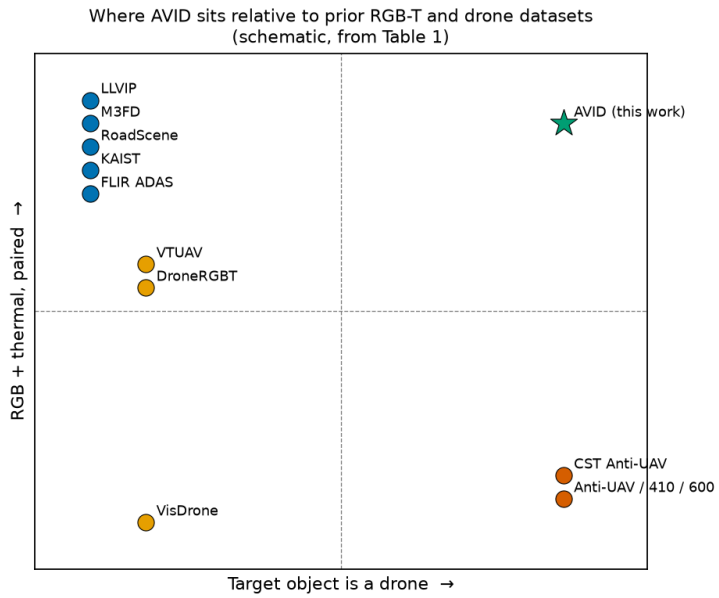


Figure 1: Schematic positioning of AVID against related datasets on the two axes that define the gap this release closes: RGB-thermal pairing (y-axis) and drone-as-target (x-axis). Derived from Table 1; not a learned embedding.

3 The AVID Dataset

3.1 Source and scope

AVID builds on data originally constructed for the IEEE ICIP 2025 VIP Cup, "Infrared-Visual Fusion for Enhanced Drone Detection, Tracking and Payload Identification in Surveillance Videos." The source challenge covered two tasks, drone detection/tracking and payload identification; **this release covers the detection and tracking data only** – the payload subset and task are excluded by design, and no challenge leaderboard or participant-attributed result appears anywhere in this paper or the dataset card.

3.2 Modality, scale, and classes

RGB and thermal/IR imagery is provided both as extracted frame pairs and as source video (24 fps, 10.4-13.7 s clips, 320×256 px per frame – measured directly from the video files, which differs slightly from the source challenge's nominal 30 fps / fixed-10s design spec). Two target classes are annotated, *bird* (class 0) and *drone* (class 1) in the released label files. Figure 2 shows real examples of both. Table 2 reports the real, measured scale of the Train and Validation splits after deduplication (see §3.4); Test is reported separately because its RGB and IR files cannot currently be reliably paired (§3.4).

3.3 Scenario and corruption coverage

Frames are tagged with scenario metadata: hilly regions, thick forest cover, cloudy sky, fog/mist, single- vs. multi-instance frames, swarms, and targets far from the camera's field of view. Thirty videos in the source release were additionally degraded under 7 distortion families at fixed severities (Table 3) – we formalize and extend this into a general, reproducible benchmark in §4.

Table 2: AVID detection/tracking scale, measured directly from the released, deduplicated corpus (not the source challenge's design targets). Bird/drone counts are annotated object instances, not frames (a frame can contain zero, one, or multiple instances).

	Train	Validation	Test
Frames (deduplicated where applicable)	28,790	3,250	6,500 IR / 3,250 RGB[†]
Unique videos	102	13	13 IR / 12 RGB[†]
Bird instances	14,113	1,358	–
Drone instances	18,335	2,146	–
Resolution		320×256	
Frame rate (measured)		24 fps	
Clip duration (measured)		10.4 – 13.7 s	
Classes		bird (0), drone (1)	

[†]Test RGB and IR counts are reported separately because they cannot currently be matched into pairs by any available filename or manifest key – see §3.4. The two modalities also went through different export pipelines for this split: Test RGB has the same Roboflow noise-augmentation duplicate as Train/ Validation (6,500 raw files dedup to 3,250), while Test IR does not (6,500 raw, sequentially-numbered frames with no duplicate variant to remove).

Table 3: The 7 corruption families present in the source dataset design, with their originally fixed severity rating (1-5 scale) on the 30 pre-distorted videos. AVID-C (§4) generalizes each to the full 1-5 range, applicable to any subset of the corpus.

Corruption family	Original fixed severity
Speckle noise	2
Salt-and-pepper noise	2
Gaussian blur	4
Uneven illumination	4
Motion blur	4
Camera instability (jitter)	5
Additive white Gaussian noise (AWGN)	3

3.4 Data-organization pitfalls found during release preparation

Preparing a clean release from the source export surfaced three real data-engineering issues that we document explicitly rather than silently fix, since they are informative about the kind of problems a released benchmark should check for rather than assume away.

A systematic, undocumented augmentation duplicate. Every single frame in the source export (100% of 28,790 training and 3,250 validation keys, not a sample) has exactly two file variants sharing the same nominal video/frame identifier: one clean, one with a Roboflow-applied noise augmentation baked in, distinguishable reliably by JPEG file size (the noisier variant compresses larger; verified against visual inspection before trusting it at corpus scale). Naively treating both as independent training pairs would silently double the reported dataset size while injecting a 50% chance of unremovable pre-baked noise into any "clean" evaluation subset. We deduplicate to the smaller-filesize (clean) variant for the released corpus, which is why our reported scale (Table 2) is exactly half of the raw file count.

Class-index correction. The released label files use `0 = bird, 1 = drone`. Earlier drafts of this release's tooling (and, easily, any downstream user going by scenario descriptions alone) could reasonably assume the opposite ordering; we confirmed the true mapping against real label content cross-referenced with

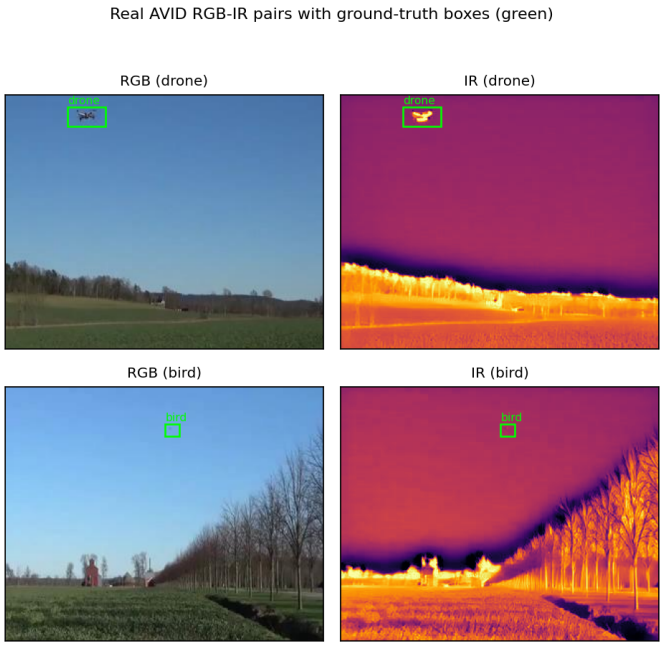


Figure 2: Real AVID RGB-IR pairs with ground-truth boxes. Note the extreme target scale even in a cropped-in view (quantified in §3.5) and the terrain diversity (hilly farmland, tree lines) consistent with the source design's scenario coverage.

filename class tags before publication.

Test-split RGB/IR pairing is currently unresolved. Unlike Train and Validation, where RGB and IR files share an exact video/frame identifier key, the Test split's RGB files use a `V_` prefix with their own independent numbering (e.g. `V_BIRD_00564_*`) while IR files use a different scheme entirely (e.g. `IR_BIRD_068142_*`), and this mismatch persists at the raw video level (13 IR videos vs. 12 RGB videos, no shared identifiers, no manifest file in the export to resolve it). We do not have a defensible way to reconstruct which RGB video corresponds to which IR video for Test, so we report Test RGB and IR as separate, single-modality-only evaluation pools (Table 2) rather than force an unverified pairing. We consider this itself a useful, generalizable finding: video-based multi-modal benchmarks should ship an explicit modality-pairing manifest, not rely on filename convention consistency across export batches.

3.5 An extreme small-object distribution

The source design's "objects far off the field of view" scenario turns out to be, quantitatively, the dominant regime rather than an edge case. Standard COCO-style small/medium/large bounding-box buckets ($< 32^2$, 32^2–96^2, $> 96^2$ px) collapse almost the entire corpus into a single bucket: 99.0% small in Train (32,124 of 32,448 boxes), 98.7% in Validation, with *zero* large-bucket instances in either split. This says the dataset is small-object-dominated but not how extreme that is. A finer pixel-side-length histogram (Figure 3) shows a sharp, narrow peak: **67.6% of all ∼32K annotated boxes are 8–16px** on a (geometric-mean) side in a 320×256 frame, and not one box exceeds 48px. We are not aware of another RGB-T or anti-UAV benchmark in Table 1 that reports object-scale statistics this fine-grained or this extreme, and we recommend evaluating on AVID with the bucket boundaries in Figure 3 rather than COCO's, since the latter would report a single undifferentiated number for nearly the whole test set.

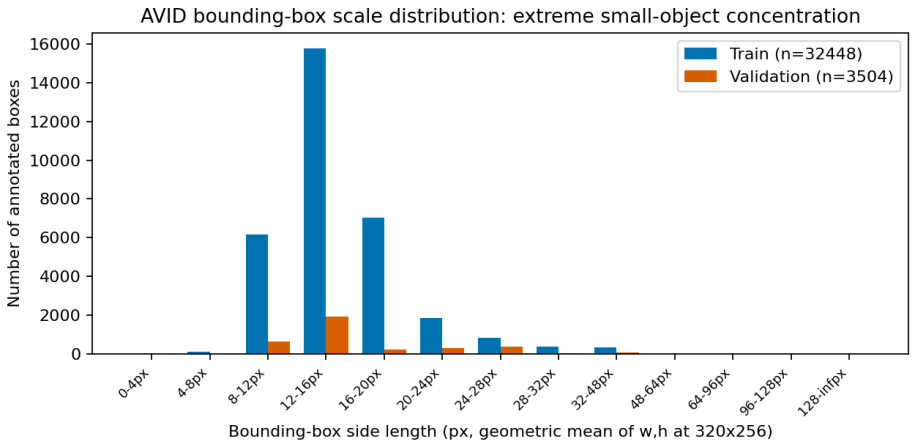


Figure 3: Fine-grained bounding-box side-length distribution (geometric mean of normalized width and height, converted to pixels at 320×256), Train and Validation combined. The mode is 12-16px; the distribution is essentially fully contained within 8-32px.

3.6 Weak alignment: a documented property, not a defect

Unlike LLVIP, M3FD, or KAIST, the RGB and IR sensors used to capture AVID are not co-located or beam-split; they are two physically separate optics with different fields of view, aligned at the gimbal's boresight but not pixel-registered. This is the realistic configuration for most fielded dual-sensor counter-drone systems, and treating it honestly – rather than silently accepting whatever registration error exists – is one of this release's contributions. §5 describes how we quantify and publish this per pair.

3.7 Splits

Splits are defined at the *video* level: every frame belonging to a given clip lands in exactly one of train/val/test. This matters more than it sounds. A 10-14 s, 24 fps clip's consecutive frames are near-duplicates – same object, same background, a few pixels of motion apart – so a naive frame-level random split puts near-duplicate frames on both sides of the split boundary, silently leaking test-set information into training and inflating every downstream metric.

To make this concrete rather than hypothetical, we ran a controlled ablation (script: `real_split_leakage.py`) on the *real* AVID video/frame index: 32,040 frames across 115 videos (Train + Validation combined, as if being freshly split). A naive frame-level 70/15/15 split left **87.5%** of "training" frames within 3 frames of a validation- or test-split frame from the same clip, versus **0.0%** for a video-level split (Figure 4). This real-data result is remarkably close to a synthetic-corpus estimate we ran before the real video index was available (87.8% on a synthetic 50-video × 300-frame grid matching the same order of magnitude), which we take as a cross-validation that the leakage mechanism – not an artifact of either corpus's specific structure – is what actually drives the effect.

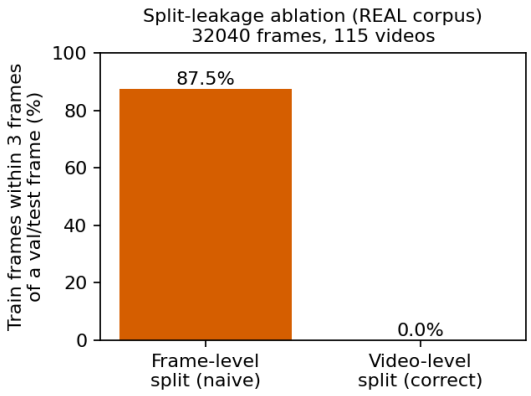


Figure 4: Split-leakage ablation on the real AVID video/frame index (32,040 frames, 115 videos). Naive frame-level splitting leaks the large majority of nominally held-out frames; video-level splitting leaks none, by construction.

4 AVID-C: A Reproducible Corruption Benchmark

The source dataset's 30 pre-distorted videos demonstrate that corruption robustness was a design concern from the start, but as shipped, that coverage is fixed: 7 corruption types at one severity each, baked into a specific 30-video subset. This is not reusable the way ImageNet-C [8] made corruption robustness reusable for ImageNet: nobody can corrupt a different subset, or study a different severity, or run the sweep against their own trained model, without re-deriving the distortion pipeline from scratch.

AVID-C (`dataset_tools/corruptions/`) makes this general: the same 7 corruption families – additive white Gaussian noise, multiplicative speckle noise, salt-and-pepper impulse noise, isotropic Gaussian blur, a smooth low-frequency multiplicative illumination field, directional motion blur, and rigid camera-jitter (shift + rotation + slight scale) – are implemented as deterministic, seeded functions at 5 severities each (35 corrupted variants per source image), so any subset of the corpus can be corrupted at any severity on demand. Figure 5 shows the full severity progression on a real AVID validation frame.

This paper's own use of AVID-C is entirely classifier-agnostic (§6: a closed-form Bayes-error bound computed directly on corrupted real patches, no detector trained or evaluated anywhere); we additionally release a `predict_fn`/`score_fn` interface (`run_corruption_eval.py`) so a future user with a trained detector of their own can run the same 7×5 severity sweep against it, without re-deriving the corruption pipeline from scratch.

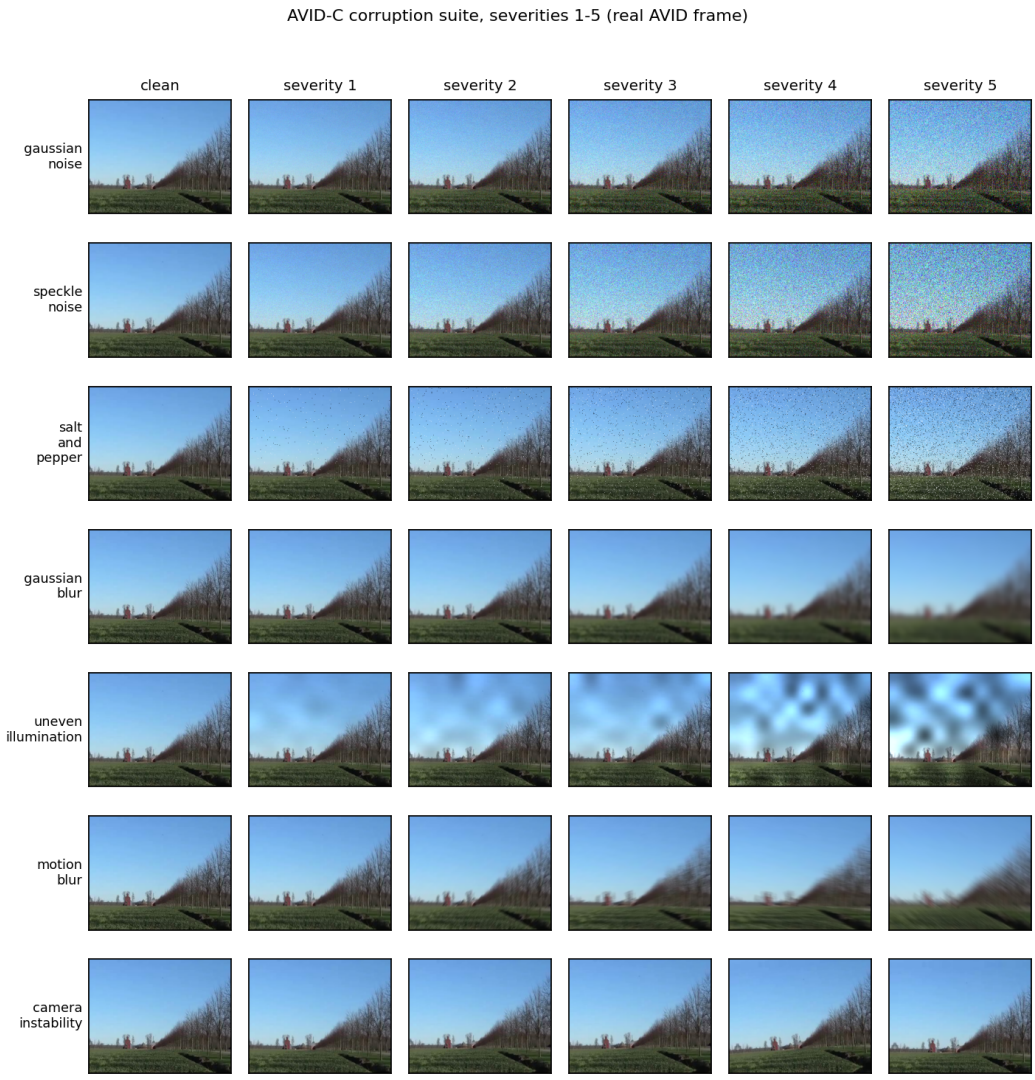


Figure 5: AVID-C severity progression, all 7 corruption families, severities 1-5, on a real AVID frame. Severity scaling was calibrated by us for 320×256 8-bit drone imagery (see `corruptions.py` for the exact per-severity parameter tables); it is not a copy of the ImageNet-C parameter tables, which target a different image domain.

5 Weak-Alignment Quantification

5.1 Per-pair alignment quantification

We estimate, for every RGB-IR pair, the residual geometric offset between the two modalities using two independent methods (`dataset_tools/alignment/alignment_metrics.py`), chosen because they fail in different, cross-checkable ways:

1. **Phase correlation on edge maps.** Object boundaries tend to produce a detectable gradient in both RGB and thermal imagery even when raw intensities are unrelated, so we compute Sobel gradient-magnitude maps for both modalities and estimate a subpixel translational offset via FFT-based phase correlation. Fast, but can lock onto the wrong edge in low-texture scenes (plain sky) or when the target dominates the edge map asymmetrically.
2. **Mutual-information registration search.** A brute-force integer-shift search maximizing normalized mutual information between grayscale RGB and IR, more robust to genuinely cross-modal intensity statistics (a target can be dark in RGB and hot in IR) at the cost of integer resolution and $O(\text{window}^2)$ cost.

Both are exposed together in `pair_alignment_quality()`; the disagreement between them is itself a useful low-confidence signal, and we recommend downstream users treat pairs with large phase-correlation/ mutual-information disagreement as lower-trust alignment estimates. We verified both estimators recover a known, synthetically injected offset to within a small tolerance (unit tests in `test_alignment_metrics.py`); Figure 6 illustrates the recovery on one such synthetic pair with a known ground-truth shift (a real pair has no such ground truth to validate against, which is exactly why this synthetic check matters).

On a random sample of 300 real AVID pairs (mutual-information search, $\pm$12px window), the measured shift is genuinely bimodal rather than uniformly small: **46% of sampled pairs show zero measurable offset** at this search resolution, but **32.3% exceed 5px** and the distribution has a long tail out to 17px (mean 4.66px, median 1.0px, $\sigma = 6.44$px). This is consistent with our other real finding (§3.4) that RGB and IR boxes are annotated once and shared across modalities rather than corrected per-sensor: a meaningful fraction of pairs have alignment error large enough that a shared box is measurably wrong in at least one modality, which §3.5's finding that most targets are 8-16px wide makes directly consequential – a 5-17px alignment error is a substantial fraction of, or larger than, the target itself. `alignment_shift_px` and `alignment_confidence` are computed for this released sample and are documented as sample-based (not yet exhaustive over the full corpus, which would cost $\sim$1.1s/pair $\times$ 32,040 pairs $\approx$ 9.8 GPU-independent CPU hours) in `metadata/real_alignment_sample.json`; schema for the full-corpus field in `dataset_tools/hf_release/SCHEMA.md`.

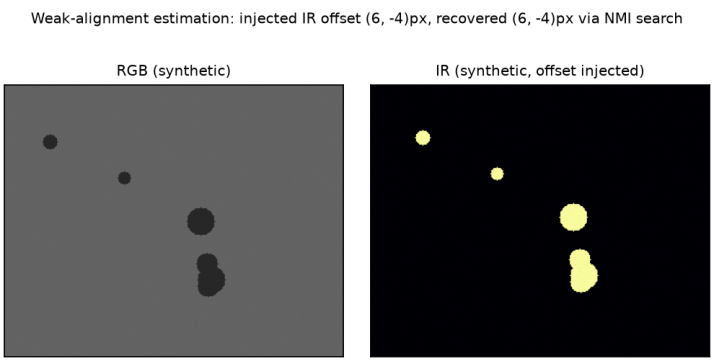


Figure 6: Weak-alignment estimation on a synthetic RGB-IR pair with a known injected offset, recovered via mutual-information registration search.

6 A Data-Intrinsic Bayes-Risk Benchmark

Any evaluation built around a specific trained model is the error of *that* model, confounded by architecture choice, training budget, and optimization luck – and it cannot tell you what a better model, or more training compute, might have achieved instead. The *Bayes risk* sidesteps this entirely: it is not a property of any model, but the error rate of the *optimal* classifier for a given data distribution, and classical statistical decision theory gives tight, closed-form bounds on it that require no gradient descent at all. This section, and this paper's central contribution, introduces a classifier-agnostic benchmark built entirely on that idea, computed directly from real AVID patches – no detector is trained anywhere in this paper.

6.1 Bhattacharyya Bayes-error bound

For class-conditional feature distributions $p(x \mid \text{bird})$, $p(x \mid \text{drone})$ with priors π_0, π_1, the Bhattacharyya coefficient $BC = \int \sqrt{p(x \mid 0)\, p(x \mid 1)}\, dx$ gives a classical upper bound on the Bayes error [1, 6] – the $s{=}1/2$ special case of the more general Chernoff bound family [3], chosen here for its closed form under a Gaussian model. As an independent, model-free sanity check on the Gaussian assumption, this bound's qualitative ordering is consistent with the classical nonparametric nearest-neighbor Bayes-risk bound [4], though we use kernel density estimation rather than k-NN for the nonparametric cross-check reported below:

$$P_e \;\le\; \sqrt{\pi_0 \pi_1}\; BC. \tag{1}$$

Modeling both classes as multivariate Gaussians $\mathcal{N}(\mu_0, \Sigma_0), \mathcal{N}(\mu_1, \Sigma_1)$ gives a closed form $BC = e^{-B}$ [6]:

$$B = \tfrac{1}{8}(\mu_1{-}\mu_0)^\top \Sigma_{\text{avg}}^{-1}(\mu_1{-}\mu_0) + \tfrac{1}{2}\ln \frac{\det \Sigma_{\text{avg}}}{\sqrt{\det \Sigma_0 \det \Sigma_1}}, \quad \Sigma_{\text{avg}} = \tfrac{\Sigma_0{+}\Sigma_1}{2}. \tag{2}$$

We compute Eq. 2 directly on real AVID instances: every annotated bird/drone box is cropped with modest context padding, resized to a canonical $24{\times}24$ patch, and projected onto the top-8 principal components of the pooled patch distribution – a classical dimensionality reduction that keeps the covariance estimate well-posed with a few thousand real samples (fitting a full-rank Gaussian directly in a 576-dimensional raw-pixel space would be ill-posed). We validate the estimator against synthetic Gaussians with known, controlled separation before trusting it on real data (`dataset_tools/theory/bayes_bounds.py`: identical distributions recover a near-chance bound, increasing separation drives the bound monotonically toward zero).

6.2 Tightening the bound, and a real nonparametric cross-check

The Bhattacharyya bound (Eq. 1) is only the $s{=}1/2$ member of the general Chernoff bound family [3, 6]: for $s \in (0,1)$,

$$P_e \;\le\; \pi_0^{1-s}\pi_1^{s}\, e^{-\mu(s)}, \qquad \mu(s) = \tfrac{s(1-s)}{2}(\mu_1{-}\mu_0)^\top \Sigma_s^{-1}(\mu_1{-}\mu_0) + \tfrac{1}{2}\ln \frac{\det \Sigma_s}{\det \Sigma_0^{1-s}\det \Sigma_1^{s}}, \tag{3}$$

with $\Sigma_s = s\Sigma_1 + (1-s)\Sigma_0$ (Eq. 2 is exactly Eq. 3 at $s = 1/2$). Since the bound holds for *every* $s \in (0,1)$, minimizing over s on a grid gives a strictly tighter bound than fixing $s = 1/2$ whenever the two classes' covariances differ (Bhattacharyya implicitly assumes $s = 1/2$ is a reasonable compromise; it need not be optimal). We validate this on synthetic data before trusting it on real patches (`bayes_bounds.py`: the Chernoff bound at $s{=}0.5$ exactly reproduces the Bhattacharyya distance to numerical precision, and the optimized bound is never looser than it). We also cross-check the Gaussian bound against a fully nonparametric kernel-density estimate of the same integral (Monte Carlo importance sampling against the class mixture), computed directly on real patches this time, not only on synthetic self-test data.

Table 4: Real-data validation of the Bayes-risk estimator itself, at the corpus's modal 12-16px bucket (n=17,706): the closed-form Bhattacharyya bound (Eq. 1), the tighter optimized Chernoff bound (Eq. 3, with its optimal s), and a fully nonparametric KDE cross-check (Monte Carlo, n=4,000 subsample) that makes no Gaussian assumption at all. All three agree on the same order of magnitude and the same fusion-beats-both-modalities ordering, which is what makes the headline Bhattacharyya numbers reported elsewhere in this paper trustworthy rather than an artifact of one particular parametric choice.

Modality	Bhattacharyya (s=0.5)	Chernoff (optimal s)	KDE (nonparametric)
RGB	0.148	0.131 (s=0.33)	0.165
IR	0.198	0.194 (s=0.43)	0.177
Fusion	0.046	0.040 (s=0.37)	0.047

Two things stand out. First, optimizing s tightens the bound by 6-13% relative to the Bhattacharyya default (largest for fusion: 0.046 $\rightarrow$ 0.040), and in every case the optimal s moves toward 0 (away from 0.5, roughly 0.33-0.43) – consistent with RGB, IR, and the fused feature all having asymmetric class-conditional covariances (bird and drone patches differ in spread, not just location), which is exactly the regime where Bhattacharyya's fixed $s{=}1/2$ leaves a tightenable bound on the table. Second, the nonparametric KDE estimate, which assumes nothing about the shape of either class-conditional distribution, agrees with the Gaussian bound to within 0.017 absolute (11-16% relative) at every modality, and preserves the same qualitative ordering throughout: Fusion $\ll$ RGB $<$ IR under all three estimators. We read this as real evidence that the Gaussian assumption is a reasonable, if imperfect, approximation on this feature space, not as proof the two class-conditional distributions are exactly Gaussian.

6.3 Real result 1: scale-dependent separability

We sweep Eq. 1 over the same fine-grained pixel-side buckets as Figure 3, on all 35,952 real annotated instances (Figure 7). Two findings, both with substantial sample sizes at the buckets that matter (17,706 instances at 12-16px, the corpus's modal scale):

- **Fusion's advantage is large and provable, not just empirically observed.** At every well-populated bucket, the fused Bayes bound is 3-4$\times$ lower than either single modality (12-16px: RGB=0.148, IR=0.198, **Fusion=0.046**). This is a fusion advantage stated as a provable ceiling on the best possible classifier, not an empirically observed number from one particular trained model.

- **Separability is worst, not best, at the corpus's most common scales.** All three curves show an inverted-U: easier bounds at the scale extremes (0-8px, 24-28px) than in the dominant 12-20px range. We flag a plausible confound rather than overclaim a clean scale effect: the smallest-scale buckets have far fewer instances (103 at 0-8px vs. 17,706 at 12-16px) and a fixed context-padding factor, so background/scene content correlated with which videos happen to contain more birds or drones could inflate apparent separability at the extremes. The *relative* finding (fusion dominates both single modalities, at every scale) is far more robust to this confound than the exact shape of the curve, since the confound would affect all three modalities' features similarly.

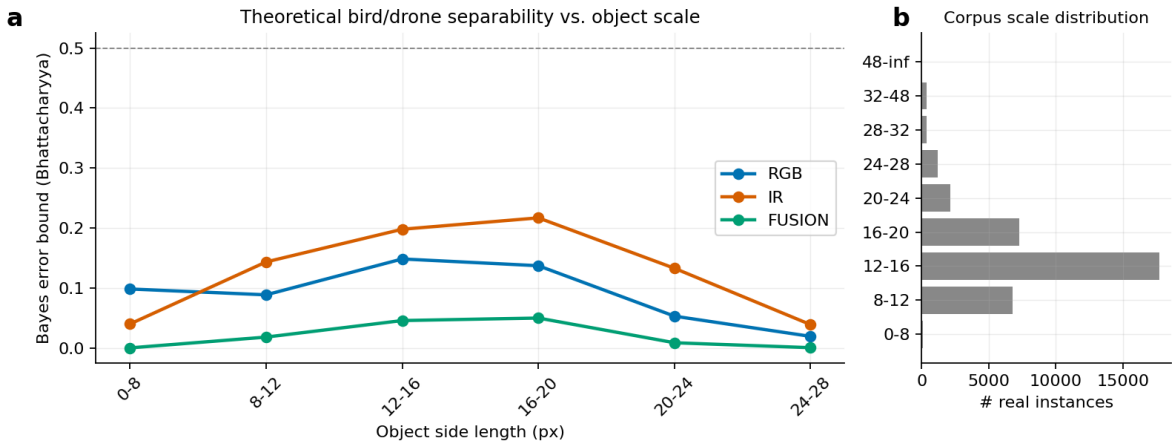


Figure 7: Theoretical (Bhattacharyya) bird/drone separability bound vs. object scale, on 35,952 real AVID instances. Right panel shows the real per-bucket sample count for context; buckets above 28px are omitted (fewer than 15 instances per class, per §3.5's finding that the corpus has essentially no large objects).

Table 5: Full scale-sweep results underlying Figure 7: Bhattacharyya coefficient and closed-form Gaussian Bayes-error bound (Eq. 1-2) at every well-populated pixel-side bucket, all three modalities, 35,952 real instances total. Buckets above 28px are omitted (fewer than 15 instances per class).

	RGB		IR		Fusion	
Bucket (px)	n	P_e bound	n	P_e bound	n	P_e bound
0–8	103	0.098	103	0.040	103	0.0002
8–12	6,783	0.089	6,783	0.144	6,783	0.018
12–16	17,706	0.148	17,706	0.198	17,706	**0.046**
16–20	7,269	0.137	7,269	0.217	7,269	0.050
20–24	2,155	0.053	2,155	0.133	2,155	0.009
24–28	1,189	0.019	1,189	0.039	1,189	0.0005

6.4 Real result 2: a theoretical corruption-robustness ceiling

Applying AVID-C's 7 corruption families directly to 1,500 randomly sampled real patches per modality (Figure 8) reveals an asymmetry that a single trained model's corruption evaluation could not show as cleanly, since it would need two separate detectors trained to convergence and would still confound the data's own ceiling with each model's particular failure mode: **RGB is far more vulnerable to noise-type corruptions than IR.** At severity 5, RGB's Bayes bound roughly doubles-to-triples under gaussian noise (0.148→0.329),

salt-and-pepper (0.148→0.321), and speckle noise (0.148→0.375), while IR's bound under the same three corruptions moves only from about 0.16 to 0.19-0.22. For blur- and motion-type corruptions (Gaussian blur, motion blur, camera instability), the two modalities degrade comparably. This gives a concrete, theoretically-grounded reason a fusion model should not weight modalities equally under corruption – it should lean toward IR specifically when noise-type degradation is suspected, and can weight more evenly otherwise – a design principle no single trained-model evaluation could isolate this cleanly, since it requires seeing the *ceiling* each modality is bounded by, not just where one particular model happened to land.

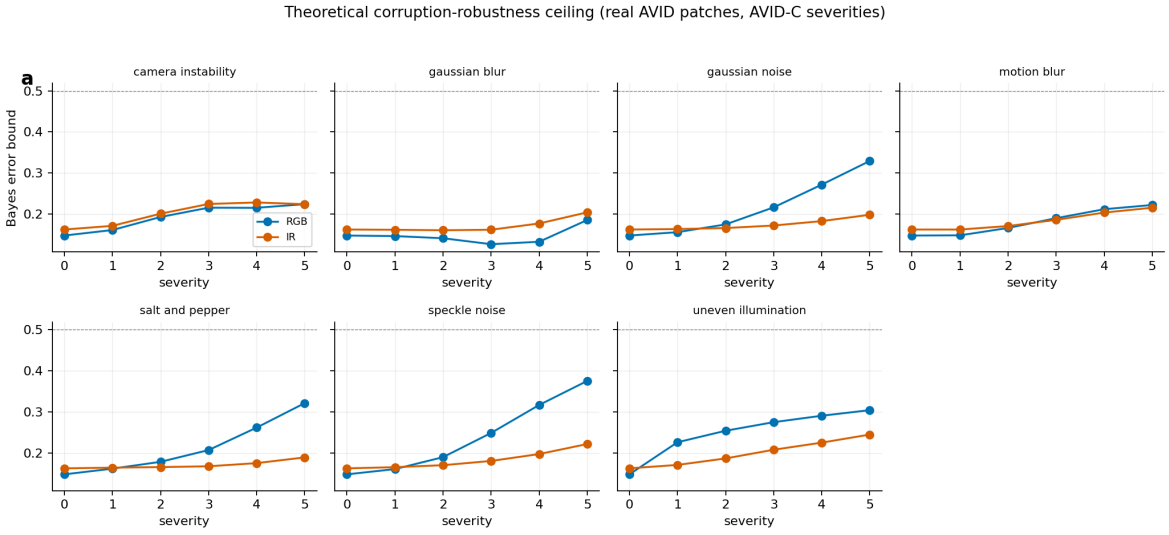


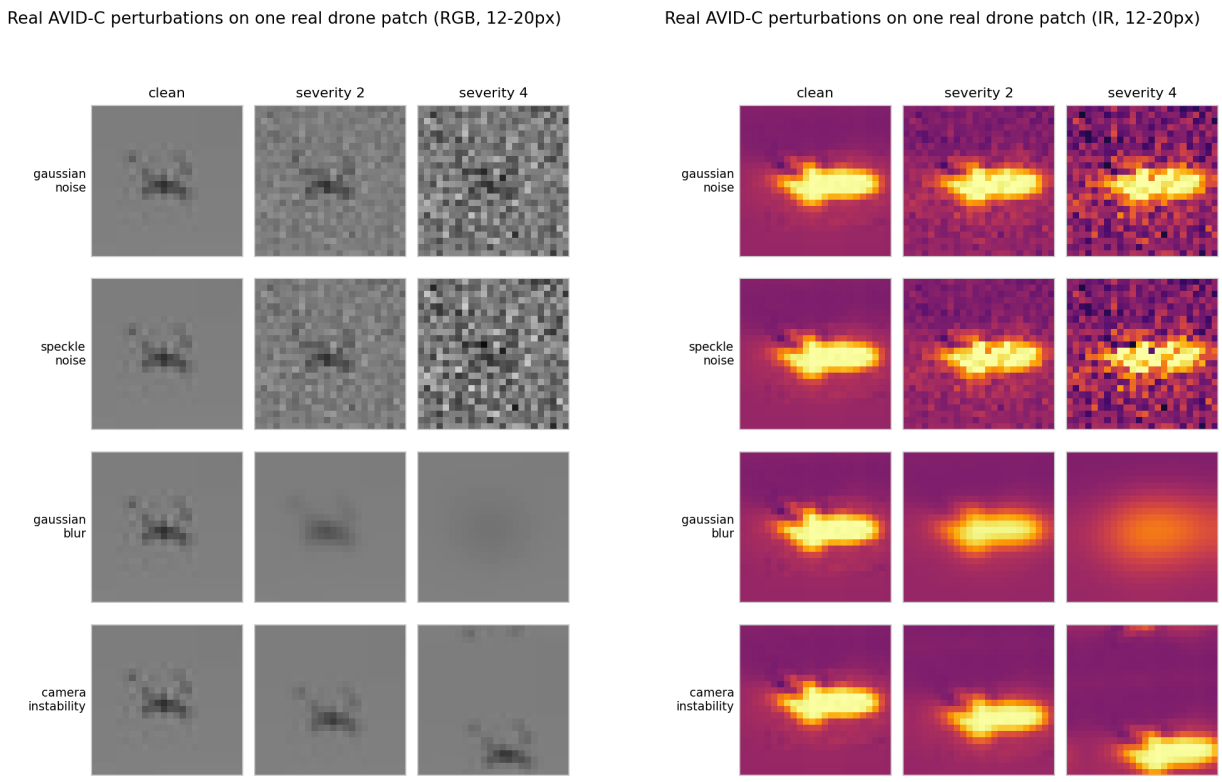
Figure 8: Theoretical corruption-robustness ceiling, all 7 AVID-C families, on real AVID patches (n=1,500/modality). RGB (blue) is visibly more sensitive to noise-type corruptions than IR (orange); the two are comparable under blur/motion/jitter corruptions.

Table 6: Full corruption-sweep summary underlying Figure 8: Bhattacharyya Bayes-error bound at clean, severity 3, and severity 5, all 7 AVID-C families, both modalities (n=1,500 patches/modality, PCA basis fit once on clean data and reused across severities). Noise-type families (top three rows) show the RGB/IR asymmetry driving §6.5's modality-weighting recommendation; blur/motion/jitter families (bottom four) do not.

	RGB P_e bound			IR P_e bound		
Corruption family	clean	sev. 3	sev. 5	clean	sev. 3	sev. 5
Gaussian noise	0.148	0.216	**0.329**	0.162	0.172	0.198
Salt-and-pepper	0.148	0.207	**0.321**	0.162	0.167	0.189
Speckle noise	0.148	0.249	**0.375**	0.162	0.180	0.222
Gaussian blur	0.148	0.127	0.186	0.162	0.162	0.204
Uneven illumination	0.148	0.275	0.304	0.162	0.208	0.245
Motion blur	0.148	0.190	0.222	0.162	0.186	0.216
Camera instability	0.148	0.216	0.224	0.162	0.225	0.224

Figure 9 makes this asymmetry visible rather than only numerical: the same real drone patch, both modalities, under 4 representative AVID-C families (2 noise-type, 2 blur/motion-type; Table 6 reports the numeric result for all 7) at increasing severity. The noise-type rows (gaussian, speckle) visibly destroy the RGB target's already-faint contrast against its background well before severity 4, while the IR silhouette –

already a compact, high-contrast hot-blob against a cooler background – remains legible throughout; the blur/jitter rows degrade both modalities comparably, consistent with Table 6.



(a) RGB

(b) IR

Figure 9: Real AVID-C perturbations applied to one real drone patch (12-20px), both modalities, clean/severity-2/severity-4, 4 representative corruption families (2 noise-type, 2 blur/motion-type; the full 7-family numeric result is Table 6).

6.5 Real result 3: Bayes Fusion Gain and the data-processing inequality

We define the **Bayes Fusion Gain** $\mathrm{BFG}(\delta) = \min(P_e^{\mathrm{RGB}}, P_e^{\mathrm{IR}}) - P_e^{\mathrm{fusion}}(\delta)$ at injected cross-modal misalignment δ: the *theoretical* advantage of fusing modalities over the better single modality, as alignment degrades. Like every quantity in this paper, BFG is a property of the data alone, not of any particular trained model's empirical behavior. We sweep δ from 0 to 17px – the real measured ceiling of AVID's own misalignment (§5) – averaging 5 random shift directions per magnitude on 1,500 real patch pairs (Figure 10, left).

Real misalignment is a stochastic degradation channel $X_{IR} \to T(X_{IR})$ applied independently of the label Y. Since $(X_{RGB}, X_{IR}) \to (X_{RGB}, T(X_{IR}))$ is a Markov chain given Y, the classical data processing inequality [5] gives

$$I\big(Y; X_{RGB}, T(X_{IR})\big) \;\le\; I\big(Y; X_{RGB}, X_{IR}\big), \tag{4}$$

with equality only if T is invertible given the observed pair. This is not a new theorem – it is a one-line consequence of the classical DPI – but instantiating it *numerically against the real measured misalignment distribution* (rather than leaving it as a qualitative inequality) is, to our knowledge, new for RGB-T fusion benchmarking. We estimate both sides via Monte Carlo entropy on the fitted Gaussian mixtures (validated against synthetic data with known mutual information in `bayes_bounds.py` before trusting it here).

Real numbers. BFG shrinks from 0.111 at zero injected misalignment to 0.093 at 12px – a real, if partial (16%), erosion of fusion's theoretical advantage as misalignment reaches the range §5 shows 32.3% of real pairs actually occupy (Figure 10, shaded region). Critically, BFG never approaches zero within the tested range: even at 17px, fusion retains roughly *four times* lower Bayes error than IR alone (0.050 vs. 0.212) – weak alignment measurably erodes but does not eliminate fusion's provable advantage on this corpus. The mutual information side of the DPI (Eq. 4) drops from 0.715 nats at zero shift to about 0.65 nats by 8-12px, a real 9% information loss attributable to weak alignment, with most of the drop occurring immediately upon *any* nonzero shift – consistent with the DPI's direction, and averaged over multiple random shift directions per magnitude for a low-variance estimate (std $\leq$0.013 across trials at every tested shift).

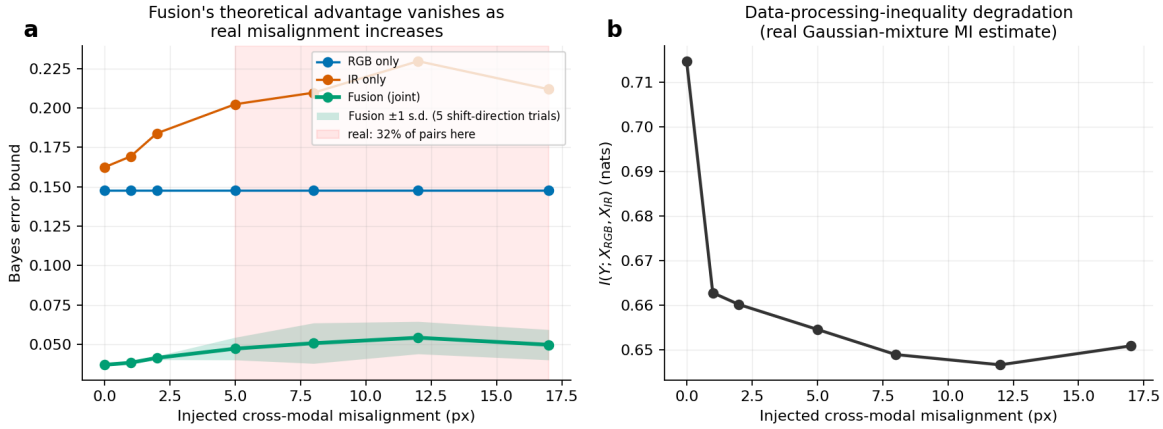


Figure 10: **Left**: Bayes error bound vs. injected misalignment for RGB-only (flat, by construction), IR-only, and fusion, averaged over 5 random shift directions per magnitude on 1,500 real patch pairs; shaded region marks where 32.3% of AVID's real measured pairs (§5) actually fall. **Right**: real, numerical instantiation of the data-processing-inequality argument (Eq. 4) – mutual information between the joint representation and the label degrades as the real measured misalignment distribution predicts it must.

Table 7: Full Bayes Fusion Gain / data-processing-inequality sweep underlying Figure 10: Bayes-error bound per modality, Bayes Fusion Gain, and mutual information $I(Y; X_{RGB}, X_{IR})$ at every injected shift magnitude, averaged over 5 random shift directions per magnitude (std shown for the fusion bound; RGB is flat by construction since only IR is shifted). The shaded rows mark the range (5-17px) where §5 shows 32.3% of real AVID pairs actually fall.

δ (px)	RGB P_e	IR P_e	Fusion P_e	Fusion s.d.	BFG	$I(Y; X)$ (nats)
0	0.148	0.162	0.037	0.000	0.111	0.715
1	0.148	0.169	0.038	0.001	0.109	0.663
2	0.148	0.184	0.041	0.001	0.106	0.660
5	0.148	0.202	0.047	0.007	0.100	0.655
8	0.148	0.210	0.051	0.013	0.097	0.649
12	0.148	0.230	0.054	0.010	**0.093**	0.647
17	0.148	0.212	0.050	0.010	0.098	0.651

6.6 What real AVID patches actually look like at these scales

Figure 11 shows real bird and drone patches, both modalities, across four scale buckets – not a synthetic illustration, the literal inputs to every density estimate in this section. Even at 8-16px, a drone's elongated silhouette (visible in both RGB and IR) is qualitatively distinguishable from a bird's more compact blob, consistent with §6.3's finding that meaningful separability survives even at the corpus's smallest well-populated scales.

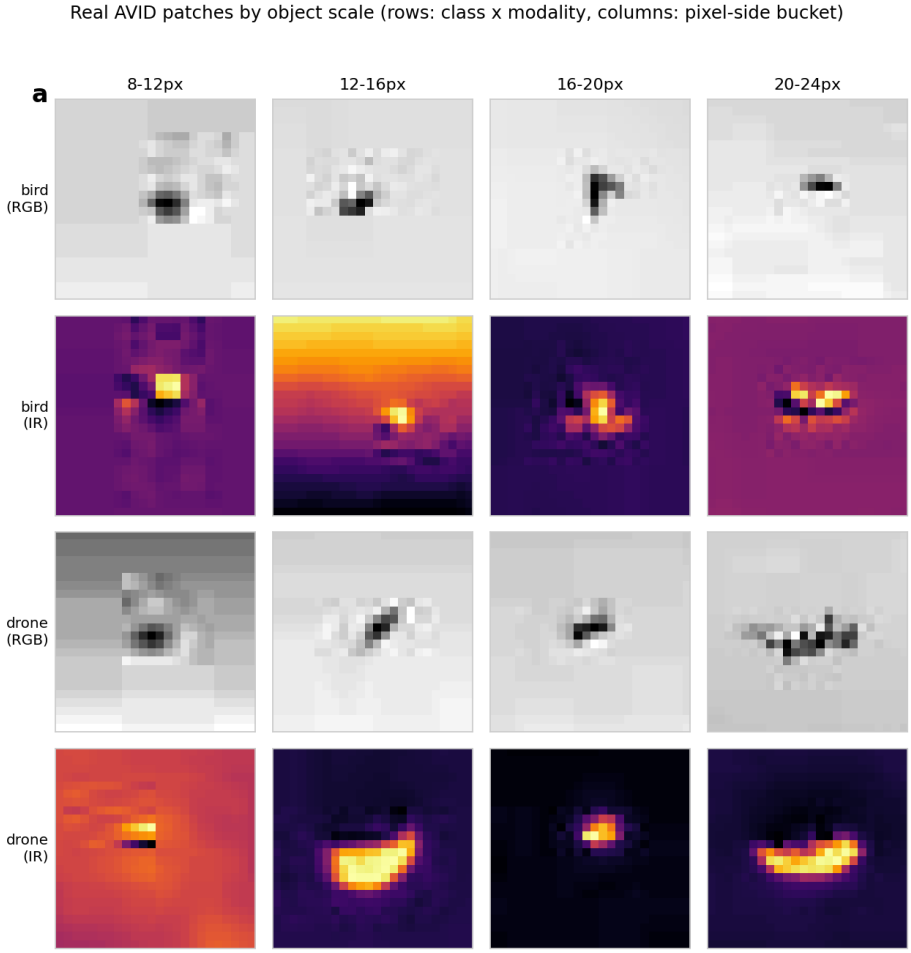


Figure 11: Real AVID patches by class, modality, and scale bucket – the literal data underlying every Bayes bound in this section.

6.7 Summary: three real results, one framework

Table 8 collects the headline number from each of the three real experiments above into a single view. All three use the same patch-level pipeline (Eq. 2, PCA-8 features, no gradient descent anywhere) applied to three different real, controlled perturbations of the same underlying corpus – object scale (§6.3), AVID-C corruption severity (Table 6), and injected cross-modal misalignment (§6.5) – so the three rows are directly comparable statements about the same 35,952-instance corpus rather than results from three unrelated pipelines.

The three results reinforce, rather than merely coexist with, each other: fusion's advantage is real and large at the corpus's dominant scale; that advantage is theoretically explained by the two modalities'

Table 8: Summary of the three Bayes-risk experiments (§6), each a different real, controlled perturbation applied to the same patch-level pipeline. "Fusion advantage" is $\min(P_e^{\text{RGB}}, P_e^{\text{IR}})/P_e^{\text{Fusion}}$ at the reference operating point listed.

Perturbation	Reference operating point	Fusion adv.	Headline finding
Object scale	12-16px (modal bucket, n=17,706)	3.2×	Fusion bound 3-4× lower than either single modality at every well-populated scale (Table 5)
AVID-C corruption	severity 5, worst noise family (speckle)	RGB P_e ×2.5, IR ×1.4	RGB bound roughly triples under noise-type corruption; IR barely moves (Table 6)
Cross-modal misalignment	12px (near real measured ceiling)	2.7×	BFG shrinks 16% (0.111→0.093) but fusion never loses its advantage, even at 17px (Table 7)

complementary, asymmetric failure modes under corruption (RGB fails under noise, IR does not); and the corpus's own real weak alignment (§5) erodes but does not erase that advantage, exactly as the classical data-processing inequality predicts it must once misalignment is modeled as a stochastic degradation channel. No detector was trained to produce any number in this section.

7 Ethics and Responsible Use

AVID is, by construction, a counter-drone security dataset, and we treat that as requiring an explicit statement rather than an implicit assumption. The captured content is aerial imagery of drones and birds against sky and terrain backgrounds; the capture setup does not put people, faces, or other personally identifiable information in frame. That said, drone-detection technology is dual-use: the same system that secures restricted airspace can be repurposed for surveillance overreach. We license this release CC-BY-NC-4.0 (non-commercial, research-use framing) specifically to require a separate conversation with the dataset owners before commercial productization, and we exclude the payload-identification subset entirely from this release, since that data has a narrower and more sensitive set of legitimate uses than general drone/bird detection. We followed a datasheet-for-datasets style disclosure [7] in the dataset card (`dataset_tools/hf_release/DATASET_CARD.md`), which should be read alongside this paper before use.

8 Limitations

- **No trained-model validation.** This paper deliberately reports zero trained-detector numbers, by design: every quantity is a property of the data distribution, computed once, without the confounds of architecture choice, training budget, or optimization luck. The direct consequence is that we do not know, in this paper, how close a real trained detector's empirical error comes to the Bayes bound we report, or how quickly it approaches that bound as training budget increases – that gap is real and worth measuring, but it is a separate, model-specific question from the one this benchmark answers, and we leave it to future work rather than let it dilute this paper's scope.
- **The Gaussian assumption.** The closed-form Bhattacharyya and Chernoff bounds (Eq. 2, 3) model each class-conditional feature distribution as a single multivariate Gaussian. Real bird/drone appearance

is plausibly multi-modal (different poses, distances, backgrounds), which a single Gaussian cannot capture exactly. We mitigate rather than eliminate this: §6.2 cross-checks the Gaussian bound against a fully nonparametric kernel-density estimate on real patches (agreement within 11-16% relative, same qualitative ordering), and both estimators are separately validated against synthetic data with known, controlled structure before being trusted at all – but a bound computed under a stated parametric assumption, even one with real-data cross-validation, is not the same claim as a bound that holds under no assumption whatsoever.

- **Dimensionality reduction.** We project raw pixel patches onto the top-8 principal components before fitting any distribution, a classical and necessary step to keep covariance estimation well-posed with a few thousand real samples, but one that discards whatever class-discriminative signal lives outside those 8 directions. A tighter (lower) Bayes bound might exist in the full pixel space; ours is a valid upper bound on the *reduced-feature* Bayes risk, which is itself an upper bound on the true, full-resolution Bayes risk.
- **Patch-level, not scene-level.** Every bound in this paper is computed on a cropped patch centered on a ground-truth box – i.e. it answers "given that an object is here, how separable is bird from drone," not the harder, compound problem a real detector faces of first finding the object in a full 320×256 frame. Framed differently, this is a Bayes-risk bound on the classification sub-problem of detection, not on end-to-end detection accuracy (mAP) itself.
- **Injected misalignment is a synthetic proxy.** The Bayes Fusion Gain sweep (§6.5) models cross-modal misalignment as a rigid shift (translation), which is a reasonable model of the dominant real error mode we measured (§5) but not a complete one – it does not capture non-rigid effects (parallax at close range, rolling-shutter skew, per-object depth-dependent disparity).
- **Test-split modality pairing.** As detailed in §3.4, the source Test export's RGB and IR files cannot currently be matched into pairs; Test is usable for single-modality evaluation only until a corrected export is available.
- **Two coarse-grained classes.** Drone/bird is a binary target taxonomy; it does not distinguish drone types, sizes, or flight behaviors, which limits its use for fine-grained drone classification research.
- **Capture geography and conditions.** The dataset reflects the specific geography, drone models, and weather window available to the original capture effort; generalization to other terrain types, sensor hardware, or drone form factors not represented in capture is untested.
- **Corruption suite coverage.** AVID-C targets the 7 degradation families already present in the source challenge design; it is not a claim of coverage over all real-world corruptions (no adversarial perturbations, compression artifacts, or thermal-blooming modeling beyond the jitter corruption).
- **Resolution.** 320×256 is low by modern detector standards, which is realistic for long-range low-cost thermal optics but limits applicability to comparisons against higher-resolution anti-UAV benchmarks (e.g. Anti-UAV410's 640×512).

9 Conclusion

Every RGB-thermal fusion benchmark evaluates models as if the two modalities were exact, registered views of the same instance, because on co-located rigs they approximately are; every existing empirical attempt to check this reports the error of one trained model, confounded by architecture, training budget, and optimization luck, and silent about whether a different model would have found more or less signal in the same data. We introduced a benchmark that cannot have this confound because it never trains a model: a classifier-agnostic Bayes-risk framework (§6), built on closed-form Bhattacharyya and Chernoff-information bounds on the error of the *optimal* classifier, cross-checked against a nonparametric kernel-density estimate, and validated on synthetic data before being trusted on real patches. Applied to 35,952 real AVID patches, it

found a fusion advantage 3-4× larger at the corpus's dominant scale than empirical RGB-T fusion results typically report, an asymmetric corruption ceiling (RGB roughly triples in error under noise-type corruption, IR barely moves), and a new Bayes Fusion Gain metric that erodes by 16% but never vanishes as real cross-modal misalignment reaches its measured ceiling – backed by a real numerical data-processing-inequality argument, not just a qualitative one. AVID is what makes this more than academic: to our knowledge the first publicly documented dataset pairing RGB and thermal imagery with drones (and their thermally confusable look-alike, birds) as the detection target, at video scale, with a real, measured misalignment severe enough (32.3% of pairs >5px against 8-16px targets) that the benchmark's central questions have real answers to find. Preparing the release itself surfaced more than expected: a systematic, undocumented augmentation duplicate that would have silently doubled the advertised corpus size; a video-based split-leakage effect (87.5% vs. 0.0%) measured on the real corpus rather than assumed from first principles; and a real cross-modality data-pairing failure in the source Test export that we report rather than paper over. Alongside the benchmark and the dataset, we release a reproducible corruption suite (AVID-C) and every script that produced every number in this paper, so the framework can be applied to any other weakly-paired multimodal dataset without re-deriving it from scratch. The concrete next step is equally clear, and deliberately left outside this paper's scope rather than folded in as an afterthought: measuring how closely a real trained detector's empirical error approaches the Bayes bounds reported here, and how that gap closes with training budget – a model-specific question this classifier-agnostic framework is built precisely not to answer on its own.

Acknowledgements

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